

Assignment 3 - Programming Error Analysis Sheet

Question No.	Error	Line No.	Error Type	Error Correction
1	Missing semicolon after int x = 10	8	Syntax Error	int x = 10;
1	Assignment operator = instead of comparison == in if	13	Logical Error	if(x == 10)
1	String assigned to int variable	18	Type Error	int marks = 90;
1	Undefined variable z	20	Semantic Error	int z = 0; printf("%d\n", z);
1	Lvalue required - 100 = x is invalid	22	Syntax Error	x = 100;
1	Array index out of bounds (arr[10] for size 5)	26	Runtime Error	arr[4] = 50;
1	Incompatible pointer type - float* to int*	28	Type Error	int *ptr = &x;
1	Wrong format specifier %f for int	30	Type Error	printf("%d\n", x);
1	Division by zero	35	Runtime Error	Check if b != 0 before division
1	Multi-character constant 'AB'	37	Type Error	char ch = 'A';
1	Same message in if and else (logical error)	39	Logical Error	if(x > y) printf("X is greater"); else printf("Y is greater");
1	sqrt() return value unused (warning)	44	Logical Error	int result = sqrt(25);
1	Undefined function show()	46	Semantic Error	Define show() or remove call
1	Semicolon after while makes empty loop	48	Logical Error	while(x < 15) { x++; }

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1	continue outside loop	53	Syntax Error	Remove or place inside loop
1	break outside loop/switch	55	Syntax Error	Remove or place inside loop/switch
1	Function calculate() called with 1 argument instead of 2	57	Semantic Error	<code>int result = calculate(10, 20);</code>
2	Missing semicolon after <code>int num2 = 20</code>	9	Syntax Error	<code>int num2 = 20;</code>
2	Missing closing parenthesis in if	21	Syntax Error	<code>if(choice == 1)</code>
2	Extra closing parenthesis in else if	33	Syntax Error	<code>else if(choice == 3)</code>
2	Missing closing parenthesis in for loop	50	Syntax Error	<code>for(int i=0; i<5; i++)</code>
2	Missing closing parenthesis in while	57	Syntax Error	<code>while(choice > 0)</code>
2	Missing semicolon after do-while	66	Syntax Error	<code>while(choice > 5);</code>
2	Missing colon after case 2	74	Syntax Error	<code>case 2:</code>
2	Missing closing quote in char literal	101	Syntax Error	<code>char grade = 'A';</code>
2	Missing closing bracket in array declaration	105	Syntax Error	<code>int arr[5];</code>
2	Missing closing brace for block	126	Syntax Error	Add <code>}</code> before <code>printf("Still Running")</code>
2	Missing semicolon after printf	133	Syntax Error	<code>printf("End of Program\n");</code>
3	String assigned to int variable	8	Type Error	<code>int age = 20;</code>
3	String assigned to	14	Type Error	<code>char grade =</code>

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	char (should be single quotes)			'A';
3	Wrong format specifier %d for float	16	Type Error	printf("%f\n", salary);
3	Wrong format specifier %f for int	18	Type Error	printf("%d\n", marks);
3	Undefined variable score	22	Semantic Error	int score = 0;
3	Variable x redeclared	25	Semantic Error	x = 20;
3	Lvalue required - 100 = x	27	Syntax Error	x = 100;
3	Wrong format specifier %s for int	31	Type Error	printf("%d\n", num);
3	Wrong format specifier %d for float	35	Type Error	printf("%f\n", pi);
3	Floating point array index	39	Type Error	arr[2] = 100;
3	Float assigned to char (truncation)	41	Type Error	char ch = 'A';
3	Function add() called with 1 argument instead of 2	45	Semantic Error	result = add(10, 20);
3	Function average() called with 2 arguments instead of 3	51	Semantic Error	avg = average(10, 20, 30);
3	String assigned to int	55	Type Error	int temperature = 35;
3	String assigned to float	59	Type Error	float discount = 10.5;
3	Float subtracted from int (implicit conversion)	61	Type Error	amount = amount - (int)discount;
3	Array name cannot be assigned	65	Syntax Error	strcpy(name, "Robil");

Question No.	Error	Line No.	Error Type	Error Correction
	directly			
3	Undefined variable totalMarks	71	Semantic Error	int totalMarks = 0;
3	Wrong format specifier %c for int	75	Type Error	printf("%d\n", number);
3	Wrong format specifier %d for float	79	Type Error	printf("%f\n", rate);
3	Function add() called with 3 arguments instead of 2	86	Semantic Error	sum = add(a, b);
3	Float result stored in int (truncation)	92	Type Error	float area; area = 3.14 * radius * radius;
3	Assignment = instead of comparison == in if	100	Logical Error	if(studentMarks == 100)
3	String assigned to int	105	Type Error	int size = 10;
3	Wrong format specifier %d for float	111	Type Error	printf("%f\n", percentage);
3	String used as array index	115	Type Error	arr2[5] = 100;
3	String literal in case label	125	Type Error	case 2:
3	Undefined function multiply()	132	Semantic Error	Define multiply() or remove call
3	Wrong format specifier %d for float	141	Type Error	printf("%f\n", areaRect);
3	Float returned but stored in int	151	Type Error	float avgMarks; avgMarks = average(...);
3	Undefined	157	Semantic Error	int value = 0;

Question No.	Error	Line No.	Error Type	Error Correction
	variable value			
3	String assigned to int	159	Type Error	char employeeId[] = "EMP101";
4	Function declared but not defined: calculateSalary	3	Semantic Error	Define calculateSalary()
4	Function declared but not defined: printStudentDetails	6	Semantic Error	Define printStudentDetails()
4	Function declared but not defined: showResult	9	Semantic Error	Define showResult()
4	Function declared but not defined: calculateAverage	10	Semantic Error	Define calculateAverage()
4	Extern variable totalStudents declared but not defined (extern needs definition elsewhere)	12	Semantic Error	Remove extern or define in another file
4	Extern variable companyProfit declared but not defined	13	Semantic Error	Remove extern or define in another file
4	Undefined function generateReport()	47	Semantic Error	Define generateReport()
4	Undefined function displayEmployee()	49	Semantic Error	Define displayEmployee()
4	Undefined function processData()	51	Semantic Error	Define processData()
4	Undefined function saveRecord()	53	Semantic Error	Define saveRecord()
4	Undefined function loadRecord()	55	Semantic Error	Define loadRecord()
4	Undefined function	57	Semantic Error	Define printInvoice

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	printInvoice()			()
4	Undefined function calculateTax()	59	Semantic Error	Define calculateTax()
4	Undefined function validateUser()	61	Semantic Error	Define validateUser()
4	Undefined function sendNotification()	63	Semantic Error	Define sendNotification()
4	Undefined function generateCertificate()	65	Semantic Error	Define generateCertificate()
5	Division by zero	10	Runtime Error	Check if b != 0 before division
5	Dereferencing NULL pointer	13	Runtime Error	if(ptr1 != NULL) *ptr1 = 50;
5	Array index out of bounds	18	Runtime Error	arr[4] = 100;
5	Dereferencing uninitialized pointer	21	Runtime Error	int *ptr2 = malloc(sizeof(int)); *ptr2 = 200;
5	Buffer overflow - strcpy into small array	25	Runtime Error	char name[20]; strcpy(name, "Programming");
5	Dereferencing uninitialized pointer	29	Runtime Error	int *ptr3 = malloc(sizeof(int)); printf("%d\n", *ptr3);
5	File not found - fopen returns NULL	33	Runtime Error	if(fp != NULL) { fscanf(...); fclose(fp); }
5	Buffer overflow in malloc'd memory	38	Runtime Error	char *ptr4 = malloc(20); strcpy(ptr4, "Computer

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				Science");
5	Use after free	43	Runtime Error	Don't access memory after free
5	Double free	48	Runtime Error	Free only once
5	Modulo by zero	53	Runtime Error	Check if y != 0 before x % y
5	Buffer overflow - strcat into small array	57	Runtime Error	char str1[30]; strcat(str1, "WorldProgramming");
5	Array out of bounds in malloc'd memory	62	Runtime Error	numbers = malloc(10 * sizeof(int));
5	2D array index out of bounds	68	Runtime Error	matrix[2][2] = 100;
5	Dereferencing NULL pointer	71	Runtime Error	if(ptr6 != NULL) printf("%d\n", ptr6[0]);
5	Using gets() - unsafe function	75	Runtime Error	Use fgets(password, sizeof(password), stdin);
5	Use after free	80	Runtime Error	Don't access ptr7 after free
5	Using uninitialized variable age	84	Runtime Error	int age = 0;
5	Negative array index	87	Runtime Error	if(index >= 0) arr[index] = 50;
5	Dereferencing uninitialized pointer	90	Runtime Error	Initialize ptr8 before dereferencing
5	Dereferencing	94	Runtime Error	if(text !=

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	NULL string pointer			NULL) printf("%s\n", text);
5	Potential memory allocation failure	99	Runtime Error	if(hugeArray != NULL) hugeArray[0] = 1;
5	Infinite loop - pointer arithmetic on NULL	104	Runtime Error	Remove infinite loop
6	Wrong logic - odd printed as even	13	Logical Error	if(num % 2 == 0) printf("Number is Even\n"); else printf("Number is Odd\n");
6	Wrong grade logic - 90+ should be A not B	23	Logical Error	if(marks >= 90) printf("Grade A\n"); else if(marks >= 80) printf("Grade B\n");
6	Incomplete leap year logic	33	Logical Error	if((year % 4 == 0 && year % 100 != 0) (year % 400 == 0))
6	Wrong logic - age > 18 should be eligible	43	Logical Error	if(age >= 18) printf("Eligible for Voting\n"); else printf("Not Eligible for Voting\n");
6	Wrong logic - smaller number logic reversed	53	Logical Error	if(a < b) printf("Smaller Number = %d\n", a); else printf("Smal

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				ler Number = %d\n", b);
6	Wrong logic for largest - using < instead of >	63	Logical Error	if(y > largest) largest = y; if(z > largest) largest = z;
6	Integer division instead of float division	73	Logical Error	average = (float)total Marks / subjects;
6	Wrong formula - area uses 2 (circumference formula)	83	Logical Error	area = 3.14 * radius * radius;
6	Wrong bonus calculation - 0.02 should be 0.20 and vice versa	93	Logical Error	if(salary > 30000) bonus = salary * 0.20; else bonus = salary * 0.02;
6	Wrong month names - case 1 is January not February	103	Logical Error	case 1: printf("Janu ary\n"); case 2: printf("Febr uary\n"); .. .
6	Wrong prime logic - isPrime set to 1 when divisible	115	Logical Error	isPrime = 0; break;
6	Wrong discount logic - adding instead of subtracting	126	Logical Error	finalAmount = purchaseAmou nt - discount;
6	Wrong sum logic - sum overwritten instead of accumulated	136	Logical Error	sum += i;
6	Wrong temperature logic - < 40 is hot not cold	146	Logical Error	if(temperatu re > 35) printf("Hot weather\n");

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				else printf("Cold Weather\n");
6	Wrong grade logic - A should be excellent not average	156	Logical Error	if(grade == 'A') printf("Exce llent Performance\ n"); else printf("Aver age Performance\ n");
6	Wrong area formula - using + instead of *	166	Logical Error	rectangleAre a = length * width;
6	Wrong pass/fail logic - >= 60 should be pass	176	Logical Error	if(percentag e >= 60) printf("Pass \n"); else printf("Fail \n");
7	Array index out of bounds - i<=5 for size 5	10	Runtime Error	for(int i=0; i<5; i++)
7	Integer division for average - should be float	20	Logical Error	average = (float)total / 5;
7	Array index out of bounds - i starts at 1, misses 0	27	Runtime Error	for(int i=0; i<10; i++) numbers[i] = i * 10;
7	Buffer overflow - strcpy into small array	37	Runtime Error	char name[20]; strcpy(name, "ComputerSci ence");
7	Array name cannot be assigned directly	42	Syntax Error	strcpy(city, "Delhi");
7	String comparison using == instead of strcmp	49	Logical Error	if(strcmp(st r1, str2) == 0)
7	Using gets() - unsafe function	56	Runtime Error	Use fgets(passwo rd, sizeof(passw

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				ord), stdin);
7	Buffer overflow - strcat may exceed array size	64	Runtime Error	char firstName[40]; strcat(firstName, lastName);
7	String not null- terminated - grade array too small	71	Runtime Error	char grade[3] = "A+";
7	Negative array index	76	Runtime Error	arr[0]
7	Buffer overflow - email array too small	87	Runtime Error	char email[30] = "student@gma il.com";
7	Array index out of bounds - scores[3] for size 3	96	Runtime Error	int scores[4];
7	String comparison using == instead of strcmp	107	Logical Error	if(strcmp(us ername, "admin") == 0)
7	Buffer overflow - strcpy into small array	114	Runtime Error	char mobile[15]; strcpy(mobil e, "98765432101 2");
7	Array index out of bounds - i<=5 for size 5	121	Runtime Error	for(int i=0; i<5; i++)
7	Array index out of bounds - course[20] for size 20	128	Runtime Error	course[0] to course[19] only
7	String not null- terminated - college array	135	Runtime Error	char college[20]; college[0]=' A';... college[5]=' \0';
7	Buffer overflow - section array too small	149	Runtime Error	char section[5] = "IT-A";

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8	Function add() called with 1 argument instead of 2	14	Semantic Error	result = add(num1, num2);
8	Function average() called with 2 arguments instead of 3	20	Semantic Error	avg = average(70, 80, 90);
8	Factorial of 0 returns 0 instead of 1	26	Logical Error	if(n == 0) return 1;
8	Swap by value - doesn't actually swap x and y in main	35	Logical Error	Use pointers: void swap(int *a, int *b)
8	Maximum returns wrong value (returns smaller)	45	Logical Error	if(a > b) return a; else return b;
8	Fibonacci recursion wrong - fibonacci(n) instead of fibonacci(n-1)	55	Logical Error	return fibonacci(n-1) + fibonacci(n-2);
8	Undefined function calculateGrade()	62	Semantic Error	Define calculateGrade()
8	Undefined function circleArea()	67	Semantic Error	Define circleArea()
8	Undefined function sumArray()	72	Semantic Error	Define sumArray()
8	Undefined function displayResult()	77	Semantic Error	Define displayResult()
8	Wrong percentage formula - integer division first	102	Logical Error	return (marks * 100) / total;
8	Function returns float but no return value	108	Semantic Error	return si;
8	Power function	121	Logical Error	for(int i=0;

Question No.	Error	Line No.	Error Type	Error Correction
	loop starts at 1 instead of 0			i<exp; i++)
8	reverseNumber doesn't return the reversed number	130	Semantic Error	return rev;
8	isPrime returns 1 for non-prime, 0 for prime (inverted)	145	Logical Error	return 0; // for non-prime, return 1 for prime
8	printArray uses <=size causing out of bounds	160	Runtime Error	for(int i=0; i<size; i++)
8	Infinite recursion - recursiveDemo(n+1) never stops	165	Runtime Error	Add base case
8	Rectangle area returns sum instead of product	175	Logical Error	return length * width;
8	cube function returns <i>nn instead of nn*n</i>	180	Logical Error	return n * n * n;
9	Dereferencing uninitialized pointer	10	Runtime Error	int *ptr1 = malloc(sizeof(int)); *ptr1 = 0;
9	Dereferencing NULL pointer	14	Runtime Error	if(ptr2 != NULL) *ptr2 = 100;
9	Pointer arithmetic out of bounds	22	Runtime Error	Don't increment ptr3 beyond array
9	Array index out of bounds via pointer	27	Runtime Error	*(ptr4 + 4)
9	Use after free	37	Runtime Error	Don't access *ptr5 after free
9	Array out of bounds in malloc'd memory	46	Runtime Error	ptr7 = malloc(10 * sizeof(int));
9	Array out of bounds - ptr8[3] for malloc(3)	55	Runtime Error	ptr8 = malloc(4 * sizeof(int))

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				;
9	Double free	62	Runtime Error	Free only once
9	Dereferencing uninitialized malloc'd memory	67	Runtime Error	*ptr10 = 0; before printf
9	Array out of bounds - numbers[5] for malloc(5)	73	Runtime Error	for(int i=0; i<5; i++)
9	Swap function swaps pointers, not values	82	Logical Error	Use int temp = *a; *a = *b; *b = temp;
9	Double pointer used incorrectly - pptr should be int**	89	Type Error	int **pptr = &value_ptr; int *value_ptr = &value;
9	Buffer overflow - strcpy into small malloc	95	Runtime Error	name = malloc(20);
9	Pointer arithmetic out of bounds after increment	102	Runtime Error	Don't access *ptr11 after ptr11++
9	Dereferencing NULL pointer	107	Runtime Error	if(ptr12 != NULL && *ptr12 > 0)
9	Array out of bounds - ptr13[100] for malloc(100)	116	Runtime Error	ptr13[99] = 500;
9	Use after free	122	Runtime Error	Don't access *ptr14 after free
9	2D array accessed via pointer with wrong offset	129	Runtime Error	Use matrix[row][col] syntax
9	Memory leak - ptr15 malloc'd twice without freeing first	135	Runtime Error	free(ptr15); before second malloc or use realloc
10	Missing semicolon after int	8	Syntax Error	int choice = 1;

Question No.	Error	Line No.	Error Type	Error Correction
	choice = 1			
10	String assigned to int	10	Type Error	int marks = 90;
10	Function add() called with 1 argument instead of 2	16	Semantic Error	result = add(a, b);
10	Assignment = instead of comparison == in if	20	Logical Error	if(a == b)
10	Undefined variable score	23	Semantic Error	int score = 0;
10	Array index out of bounds	28	Runtime Error	arr[4] = 100;
10	Negative array index	30	Runtime Error	if(index >= 0) arr[index] = 50;
10	Buffer overflow - strcpy into small array	34	Runtime Error	char name[25]; strcpy(name, "Programming Language");
10	Array name cannot be assigned directly	39	Syntax Error	strcpy(city, "Delhi");
10	String comparison using == instead of strcmp	44	Logical Error	if(strcmp(user, "admin") == 0)
10	Division by zero	50	Runtime Error	if(y != 0) printf("%d\n", x/y);
10	Dereferencing NULL pointer	54	Runtime Error	if(ptr1 != NULL) *ptr1 = 500;
10	Dereferencing uninitialized pointer	58	Runtime Error	int *ptr2 = malloc(sizeof(int));
10	Use after free	64	Runtime Error	Don't access *ptr3 after free
10	Double free	68	Runtime Error	Free only once

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10	Array out of bounds in malloc'd memory	72	Runtime Error	<code>ptr4 = malloc(5 * sizeof(int));</code>
10	Memory leak - ptr5 malloc'd twice	80	Runtime Error	<code>free(ptr5);</code> before second malloc
10	Function average() called with 2 arguments instead of 3	86	Semantic Error	<code>avg = average(70, 80, 90);</code>
10	Wrong age logic - > 18 should be eligible	94	Logical Error	<code>if(age >= 18)</code> <code>printf("Eligible\n");</code> <code>else</code> <code>printf("Not Eligible\n");</code> <code>;</code>
10	Wrong prime logic - isPrime set to 1 when divisible	104	Logical Error	<code>isPrime = 0;</code> <code>break;</code>
10	Undefined function displayReport()	116	Semantic Error	Define <code>displayReport()</code>
10	Undefined function calculateSalary()	118	Semantic Error	Define <code>calculateSalary()</code>
10	File not found - fopen returns NULL	122	Runtime Error	<code>if(fp != NULL)</code> <code>{ fscanf(...);</code> <code>fclose(fp);</code> <code>}</code>
10	Missing colon after case 1	129	Syntax Error	<code>case 1:</code>
10	Missing break in case 2	137	Logical Error	Add break after case 2
10	Multi-character constant 'AB'	142	Type Error	<code>char grade = 'A';</code>
10	Array index out of bounds - scores[3] for size 3	149	Runtime Error	<code>int scores[4];</code>

Question No.	Error	Line No.	Error Type	Error Correction
10	Array index out of bounds - i<=3 for size 3	153	Runtime Error	<code>for(int i=0; i<3; i++)</code>
10	Wrong area formula - uses 2 (circumference)	162	Logical Error	<code>area = 3.14 * radius * radius;</code>
10	Semicolon after while makes empty loop	170	Logical Error	<code>while(choice > 0) { choice--; }</code>
10	continue outside loop	174	Syntax Error	Remove or place inside loop